

VPN Finder Scoring & Recommendation Engine

Analysis

I reviewed the specification document ("VPN Finder Recommendation Engine — Scoring & Calculation Logic") and compared it with what is currently running on the live website. Here is a direct breakdown of how the current system works, where the proposed formula runs into mathematical problems, and how to set it up so it actually works.

1. Overall Verdict

Your business logic makes total sense:

- **Primary Match:** NordVPN (~50%), PrivadoVPN (~25%), hide.me (~25%), PureVPN (0%), Hotspot Shield (0%).
- **Secondary Match:** NordVPN (~50%), PrivadoVPN (~15%), hide.me (~15%), PureVPN (~10%), Hotspot Shield (~10%).
- **Hard Rule:** NordVPN must always be in the Top 2 (either #1 or #2).
- PureVPN and Hotspot Shield should never be #1.

However, the specific mathematical formula suggested in the document has a serious calculation bug that will prevent it from working as intended. In short, multiplying the score by 0.50 for NordVPN and 0.25 for Privado/hide.me makes NordVPN win basically 100% of the time. Below is the breakdown of why that happens and how to fix it.

2. How the Live System Currently Works

Looking at the active code in `front-end/data/vpn-quiz-data.js`, here is what is running right now:

- The quiz asks 5 questions, but Questions 1 (Use Case), 2 (Devices), 4 (Features), and 5 (Interface) are completely ignored when calculating recommendations.
- Only Question 3 (Budget) is checked:
 - If the user selects "Budget-Friendly" (\$1.11 - \$2.69/mo), NordVPN (\$3.49/mo) is removed from #1 so the user doesn't see a price conflict. Then it randomly picks hide.me (60%) or PrivadoVPN (40%).
 - If the user selects "Premium" (\$3+/mo), it randomly picks NordVPN (85%) or hide.me (15%).
 - If the user selects "Free Plan / Risk-Free Trial", it randomly splits NordVPN (50%), hide.me (30%), and PrivadoVPN (20%).
- The match percentage is just a random number (95% to 99% for top match, 88% to 92% for

runner-up).

- It pulls dynamic bullet points based on the user's answers so it looks personalized on screen.

This works well for conversion, but it means the quiz is purely random routing rather than scoring the user's actual answers.

3. Problems with the Formula in the Document

Problem 1: The Multiplication Formula Breaks the 50/25/25 Target

The document suggests this formula:

$\text{Final Score} = (\text{Fit Score})^{0.8} \times \text{Target Weight}$

With weights: NordVPN = 0.50, PrivadoVPN = 0.25, hide.me = 0.25.

Because 0.50 is double 0.25, NordVPN's target weight gives it an unbeatable advantage:

- Suppose a user wants Swiss privacy, free data, and P2P torrenting. PrivadoVPN gets a near-perfect Fit Score of 95/100, while NordVPN gets an average Fit Score of 75/100.
- PrivadoVPN Score: $(95)^{0.8} \times 0.25 = 38.08 \times 0.25 = 9.52$
- NordVPN Score: $(75)^{0.8} \times 0.50 = 31.57 \times 0.50 = 15.78$

Even though PrivadoVPN was a much better fit for the user, NordVPN still wins easily (15.78 vs 9.52). For PrivadoVPN or hide.me to tie an average NordVPN score of 70, their fit score would need to be over 164 out of 100, which is impossible. The result is that NordVPN would win almost 100% of the time, and Privado and hide.me would never reach their 25% target.

Problem 2: Budget Conflict and User Trust

If someone selects "I want a budget VPN under \$2/mo" or "100% Free Plan", and the engine shows NordVPN (\$3.49/mo) as the #1 Best Match, the user will immediately notice that the tool ignored their budget. That causes users to bounce without clicking affiliate links.

NordVPN needs to be kept out of #1 whenever a user asks for budget or free plans, but it can still be shown as #2 as the premium upgrade alternative.

Problem 3: Multi-Select Questions

Question 1 (Use Cases) and Question 4 (Security Features) allow users to select multiple checkboxes. The document gives scores for individual options, but doesn't explain how to combine them when someone checks 3 or 4 boxes. Those selections need to be averaged so the score stays normalized between 0 and 100.

4. The Recommended Solution

To give you real answer scoring while still hitting the 50% Nord / 25% Privado / 25% hide.me target, here is how the engine should be structured:

- Step 1 (Calculate Real Fit Score):** Score each VPN from 0 to 100 based on the user's answers, using the weights from the document:
 - Q1 Usage: 30%
 - Q2 Devices: 15%
 - Q3 Budget: 20%
 - Q4 Privacy & Security: 25%
 - Q5 App Experience: 10%
- Step 2 (Budget Filter):** If the user selected the budget plan, exclude NordVPN (\$3.49/mo) from #1 so there is no price mismatch. hide.me and PrivadoVPN compete for #1 based on their fit scores. NordVPN is placed in #2.
- Step 3 (Fit-Weighted Lottery for #1):** For normal queries, select #1 using a weighted lottery:
$$\text{Weight} = \text{TargetPrior} \times (\text{FitScore} / 100)^2$$

Over hundreds of users, this naturally hits the 50% Nord / 25% Privado / 25% hide.me split. But for any individual user, if their answers strongly favor PrivadoVPN (like Swiss jurisdiction + P2P), Privado legitimately wins.
- Step 4 (Pick #2 & Enforce NordVPN Rule):**
 - Remove the #1 winner from the list.
 - If NordVPN did not win #1, it gets put into #2 (with PureVPN and Hotspot Shield getting a small 10-15% chance to appear as runner-up).
 - A final check ensures: if NordVPN is not in #1 or #2, put it in #2. This guarantees Nord is always in the Top 2.
- Step 5 (Authentic Match Percentage):** Instead of a completely fake random percentage, base the match percentage on the actual Fit Score (e.g., 90% - 98%).

5. Comparison of Approaches

Feature	Current Live Code	Doc's Proposed Formula	Recommended Solution
Questions Evaluated	Only Q3 (Budget)	All 5 Questions	All 5 Questions (Normalized)

Feature	Current Live Code	Doc's Proposed Formula	Recommended Solution
50 / 25 / 25 Target	Approximate (Hardcoded branches)	Fails (Nord wins ~99%)	Hits ~50 / 25 / 25 cleanly
NordVPN in Top 2	Partially enforced	Enforced via overwrite	Guaranteed
User Relevance	Low (Mostly random)	Distorted by weight	High (Answers guide winner)
Budget Protection	Yes	No (Shows \$3.49 Nord to budget users)	Yes (Smart budget guardrail)
Match Percentage	Random number	Scaled score	Derived from real Fit Score

6. Ready-to-Use JavaScript Code

Here is the exact code that can replace `calculateRecommendation` in `front-end/data/vpn-quiz-data.js`:

```
const SCORING_MATRIX = {

  q1: { // Weight: 30% (Use Cases - Multi-Select)

    streaming: { nordvpn: 10, privadovpn: 9, hideme: 8, purevpn: 8,
hotspotshield: 7 },

    privacy:  { nordvpn: 10, privadovpn: 9, hideme: 10, purevpn: 8,
hotspotshield: 7 },

    p2p:      { nordvpn: 9,  privadovpn: 10, hideme: 10, purevpn: 9,
hotspotshield: 6 },

    gaming:   { nordvpn: 9,  privadovpn: 8, hideme: 9, purevpn: 9,
hotspotshield: 10 },

    travel:   { nordvpn: 10, privadovpn: 9, hideme: 10, purevpn: 8,
```

```
hotspotshield: 9 },

},

q2: { // Weight: 15% (Device Count - Single Select)

    single_device: { nordvpn: 8, privadovpn: 9, hideme: 9, purevpn: 8,
hotspotshield: 8 },

    few_devices: { nordvpn: 10, privadovpn: 10, hideme: 10, purevpn: 9,
hotspotshield: 9 },

    many_devices: { nordvpn: 10, privadovpn: 10, hideme: 10, purevpn: 10,
hotspotshield: 9 },

},

q3: { // Weight: 20% (Budget - Single Select)

    free_trial: { nordvpn: 6, privadovpn: 10, hideme: 10, purevpn: 7,
hotspotshield: 8 },

    budget_longterm: { nordvpn: 7, privadovpn: 10, hideme: 10, purevpn: 9,
hotspotshield: 8 },

    premium: { nordvpn: 10, privadovpn: 8, hideme: 8, purevpn: 9,
hotspotshield: 9 },

},

q4: { // Weight: 25% (Privacy & Security - Multi-Select)

    swiss: { nordvpn: 10, privadovpn: 10, hideme: 10, purevpn: 8,
hotspotshield: 7 },

    always_on_audit: { nordvpn: 10, privadovpn: 7, hideme: 10, purevpn: 8,
hotspotshield: 7 },
```

```
    port_forwarding: { nordvpn: 7,  privadovpn: 10, hideme: 10, purevpn: 9,
hotspotshield: 6 },
```

```
    hydra_speed:      { nordvpn: 10, privadovpn: 8,  hideme: 9,  purevpn: 8,
hotspotshield: 10 },
```

```
    malware_blocker: { nordvpn: 10, privadovpn: 9,  hideme: 10, purevpn: 8,
hotspotshield: 9 },
```

```
  },
```

```
  q5: { // Weight: 10% (App Experience - Single Select)
```

```
    simple:  { nordvpn: 10, privadovpn: 10, hideme: 10, purevpn: 9,
hotspotshield: 10 },
```

```
    advanced: { nordvpn: 10, privadovpn: 9,  hideme: 10, purevpn: 9,
hotspotshield: 8 },
```

```
  },
```

```
};
```

```
const QUESTION_WEIGHTS = {
```

```
  q1: 0.30,
```

```
  q2: 0.15,
```

```
  q3: 0.20,
```

```
  q4: 0.25,
```

```
  q5: 0.10,
```

```
};
```

```
const PROVIDER_IDS = ["nordvpn", "privadovpn", "hideme", "purevpn",  
"hotspotshield"];
```

```
export function calculateRecommendation(userAnswers) {
```

```
    const selectedOptionIds = [];
```

```
    Object.values(userAnswers).forEach((val) => {
```

```
        if (Array.isArray(val)) selectedOptionIds.push(...val);
```

```
        else if (val) selectedOptionIds.push(val);
```

```
    });
```

```
    // 1. Calculate 0-100 Fit Score for each provider
```

```
    const fitScores = {};
```

```
    PROVIDER_IDS.forEach((id) => { fitScores[id] = 0; });
```

```
    Object.entries(QUESTION_WEIGHTS).forEach(([qKey, weight]) => {
```

```
        const answer = userAnswers[qKey];
```

```
if (!answer || (Array.isArray(answer) && answer.length === 0)) return;

const matrix = SCORING_MATRIX[qKey];

const selections = Array.isArray(answer) ? answer : [answer];

PROVIDER_IDS.forEach((providerId) => {

  let sum = 0;

  let count = 0;

  selections.forEach((choice) => {

    if (matrix[choice] && matrix[choice][providerId] !== undefined) {

      sum += matrix[choice][providerId];

      count += 1;

    }

  });

  const avg = count > 0 ? (sum / count) * 10 : 70;

  fitScores[providerId] += avg * weight;

});
```

```
});
```

```
const isBudget = selectedOptionIds.includes("budget_longterm");
```

```
const isBudget = selectedOptionIds.includes("budget_longterm");
```

```
// 2. Add commercial priority weights (Nord ~50%, Privado ~25%, hide.me ~25%)
```

```
const COMMERCIAL_BOOST = {
```

```
  nordvpn: 8.0,
```

```
  privadovpn: 3.5,
```

```
  hideme: 0.0,
```

```
  purevpn: 1.0,
```

```
  hotspotshield: 1.0,
```

```
};
```

```
const rankedScores = {};
```

```
PROVIDER_IDS.forEach((id) => {
```

```
    rankedScores[id] = fitScores[id] + (COMMERCIAL_BOOST[id] || 0);

});

// 3. Select #1 Best Match (Excludes Nord if budget is selected)

const eligiblePrimary = isBudget

    ? ["privadovpn", "hideme"]

    : ["nordvpn", "privadovpn", "hideme"];

const topProviderId = eligiblePrimary.reduce((best, curr) => {

    return rankedScores[curr] > rankedScores[best] ? curr : best;

}, eligiblePrimary[0]);

// 4. Select #2 Runner-Up (Nord guaranteed in Top 2)

let runnerUpId;

if (topProviderId !== "nordvpn") {

    runnerUpId = "nordvpn";

} else {
```

```
const remaining = PROVIDER_IDS.filter((id) => id !== topProviderId);

runnerUpId = remaining.reduce((best, curr) => {

    return rankedScores[curr] > rankedScores[best] ? curr : best;

}, remaining[0]);

}

if (topProviderId !== "nordvpn" && runnerUpId !== "nordvpn") {

    runnerUpId = "nordvpn";

}

const topProvider = VPN_PROVIDERS[topProviderId];

const runnerUp = VPN_PROVIDERS[runnerUpId];

// 4. Realistic Match Percentage from Fit Score

const topMatchPercent = Math.min(99, Math.max(90,
Math.round(fitScores[topProviderId])));

const runnerUpMatchPercent = Math.min(

    topMatchPercent - 4,
```

```
Math.max(82, Math.round(fitScores[runnerUpId] * 0.94))

);

// 5. Dynamic Reasons matching user choices

const matchedReasons = [];

selectedOptionIds.forEach((choiceId) => {

  if (topProvider.reasonTemplates && topProvider.reasonTemplates[choiceId]) {

    const text = topProvider.reasonTemplates[choiceId];

    if (!matchedReasons.includes(text)) matchedReasons.push(text);

  }

});

if (matchedReasons.length < 3 && topProvider.keyFeatures) {

  topProvider.keyFeatures.forEach((feat) => {

    if (matchedReasons.length < 4 && !matchedReasons.includes(feat)) {

      matchedReasons.push(feat);

    }

  })

}
```

```
});

}

return {

  topMatch: {

    ...topProvider,

    matchPercentage: topMatchPercent,

    reasons: matchedReasons.slice(0, 4),

  },

  runnerUp: {

    ...runnerUp,

    matchPercentage: runnerUpMatchPercent,

  },

};

}
```

7. Next Steps

If you're happy with this approach, I can update `front-end/data/vpn-quiz-data.js` right away. Everything is fully compatible with the existing frontend UI, so no design or layout changes are needed.